

# COINS IN A TRAY

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**ABSTRACT.** For which  $n$  can coins of radii  $\frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n}$ , at least one of each, be held rigidly in a circular tray of radius 1? The asker found arrangements for  $n = 2, 3, 4$  and conjectured that there are none for  $n \geq 5$ . We settle two natural special cases. There is no rigid arrangement with every coin on the rim for any  $5 \leq n \leq 1000$ , and none whose contact graph is a triangulation for any  $5 \leq n \leq 16$ . The rim case reduces to deciding which integer relations hold among the angles  $\alpha_d = 2 \arcsin(1/\sqrt{d})$  and  $\pi$ ; we decide them exactly with prime-ideal valuations in imaginary quadratic fields, following Conway, Radin and Sadun. For  $n = 5$  the rim case needs no computer. We also show that no coin  $\frac{1}{k}$  with  $k \equiv 0, 1, 4, 5, 7, 8, 9, 10, 12, 13, 16, 17, 19, 20, 21, 22 \pmod{24}$  can touch the tray and two coins that touch each other and the tray. The general case remains open.

## 1. THE QUESTION

Dan asked on Mathematics Stack Exchange in June 2024 and on MathOverflow in July 2026 [1] for which  $n$  coins of radii  $\frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n}$ , with at least one coin of each size and no others, can be held rigidly in a circular tray of radius 1: no coin can move, except by spinning about its own centre or by all coins turning together about the centre of the tray. He found arrangements for  $n = 2$  (two half coins),  $n = 3$  (a half and four thirds) and  $n = 4$ , a ring  $\frac{1}{3}, \frac{1}{2}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}$  around the rim (Figure 1(b)), and conjectured that  $n \geq 5$  is impossible. Before reading on, try to place a  $\frac{1}{5}$  coin on the rim next to the coins of the  $n = 4$  ring.

Two answers made progress. Matthew House [3] ran randomised searches for  $5 \leq n \leq 10$  and found no rigid arrangement, only near-misses with gaps of  $10^{-7}$  to  $10^{-9}$ . The second answer [2], by an anonymous user, treated the case where every coin touches the tray and reduced it to the question: which integer relations hold among  $\pi$  and the angles  $\alpha_d$  below? This note answers that question and draws three consequences.

**Theorem 1.** *For  $5 \leq n \leq 1000$  there is no rigid arrangement in which every coin touches the tray. For  $n = 5$  the proof is a short argument by hand.*

**Theorem 2.** *For  $5 \leq n \leq 16$  there is no arrangement, rigid or not, whose contact graph, with the tray as a vertex, is a triangulation.*

**Proposition 3.** *If a coin of radius  $\frac{1}{k}$  touches the tray and two coins that touch each other and the tray, then  $k \equiv 2, 3, 6, 11, 14, 15, 18, 23 \pmod{24}$ .*

The asker's  $n = 4$  ring is not a triangulation: it leaves a gap with four sides. So the cases left open mix rim coins, interior coins and gaps that are not triangles.

## 2. RIM ARRANGEMENTS

Two coins  $\frac{1}{j}, \frac{1}{k}$  that touch the tray and each other have centres at distances  $1 - \frac{1}{j}$  and  $1 - \frac{1}{k}$  from the centre of the tray and  $\frac{1}{j} + \frac{1}{k}$  from each other. The law of cosines gives the angle  $\alpha_d$  between them at the centre:

$$(1) \quad \cos \alpha_d = 1 - \frac{2}{d}, \quad \text{equivalently} \quad \sin \frac{\alpha_d}{2} = \frac{1}{\sqrt{d}}, \quad d = (j-1)(k-1).$$

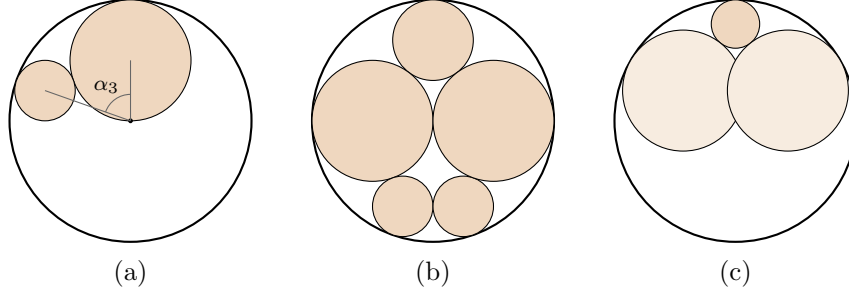
For example  $\alpha_1 = \pi$  (two half coins),  $\alpha_2 = \pi/2$ ,  $\alpha_4 = \pi/3$ , and  $\alpha_3 = \arccos \frac{1}{3}$  for the pair  $\frac{1}{2}, \frac{1}{4}$  (Figure 1(a)).

If every coin touches the tray, sort the coins by angle. The answer [2] shows that angularly adjacent coins must touch, since otherwise some coin is held only by the tray and one neighbour,

on one side, and slides. So a rim arrangement is a cyclic sequence  $\frac{1}{j_1}, \dots, \frac{1}{j_m}$  using every size, with

$$(2) \quad \sum_{i=1}^m \alpha_{d_i} = 2\pi, \quad d_i = (j_i - 1)(j_{i+1} - 1),$$

and no two coins overlapping. The asker's ring satisfies (2) because  $2\alpha_3 + \alpha_9 = \pi$ , an identity that follows from  $\cos 2\alpha_3 = -\frac{7}{9} = -\cos \alpha_9$ .



**Figure 1.** (a) Rim coins  $\frac{1}{2}$  and  $\frac{1}{4}$  subtend  $\alpha_3 = \arccos \frac{1}{3}$  at the centre. (b) The asker's  $n = 4$  ring  $\frac{1}{3}, \frac{1}{2}, \frac{1}{4}, \frac{1}{2}$ : its angles  $\alpha_2 + \alpha_3 + \alpha_9 + \alpha_3 + \alpha_2$  add up to  $2\pi$ . (c) For  $n = 5$  a  $\frac{1}{5}$  coin on the rim must sit between two half coins at  $60^\circ$  each, and then the half coins,  $120^\circ$  apart, overlap.

### 3. WHICH RELATIONS HOLD AMONG THE ANGLES

Write  $d - 1 = s^2a$  with  $a$  squarefree, and call  $a$  the *class* of  $d$ ; put  $d = 1$ , where  $\alpha_1 = \pi$ , in a class of its own. By (1),  $\tan \frac{\alpha_d}{2} = 1/(s\sqrt{a})$ , so  $\frac{\alpha_d}{2} = \frac{\pi}{2} - \arctan(s\sqrt{a})$ ; since  $\arg(1 - is\sqrt{a}) = -\arctan(s\sqrt{a})$ ,

$$(3) \quad \alpha_d = \pi + 2 \arg \beta_d, \quad \beta_d = 1 - s\sqrt{-a} \in \mathbb{Q}(\sqrt{-a}).$$

Two facts decide every relation.

*Splitting.* If  $\sum_d m_d \alpha_d$  is a rational multiple of  $\pi$  (with  $m_d \in \mathbb{Z}$ ), then so is the partial sum over each class. This is the Splitting Theorem of Conway, Radin and Sadun [4] for angles whose squared trigonometric functions are rational, which the  $\alpha_d$  are.

*Within a class.* Let  $K = \mathbb{Q}(\sqrt{-a})$  and  $\gamma = \prod_d \beta_d^{m_d}$ . Then  $\sum m_d \arg \beta_d$  is a rational multiple of  $\pi$  exactly when  $\gamma/\bar{\gamma}$  is a root of unity, that is, when the principal ideals  $(\gamma)$  and  $(\bar{\gamma})$  coincide. Primes of  $K$  that are ramified or inert equal their conjugates, so this happens exactly when, for every rational prime  $p$  that splits as  $p\bar{p}$ ,

$$(4) \quad \sum_d m_d (v_p(\beta_d) - v_{\bar{p}}(\beta_d)) = 0.$$

Each difference is computed from the norm  $1 + s^2a$  of  $\beta_d$ : if  $p^e$  exactly divides it and  $\rho$  is a  $p$ -adic square root of  $-a$ , then  $v_p(\beta_d) = v_p(1 - s\rho)$  and  $v_{\bar{p}}(\beta_d) = v_p(1 + s\rho)$ , and these add up to  $e$ . When (4) holds,  $2 \arg \gamma$  is a multiple of  $2\pi/w$ , where  $w \in \{2, 4, 6\}$  is the number of roots of unity in  $K$ ; so the class sum is an exact multiple of  $\pi/3$  or  $\pi/2$ , and a floating-point evaluation tells which.

For instance  $d = 3$  and  $d = 9$  share the class  $a = 2$ , with  $\beta_3 = 1 - \sqrt{-2}$  of norm 3 and  $\beta_9 = 1 - 2\sqrt{-2}$  of norm 9. At the split prime 3 their valuation differences are 1 and  $-2$ , so  $2\alpha_3 + \alpha_9$  satisfies (4), as the asker's ring requires.

A class containing a single  $d$  gives the simplest consequence: then  $m_d \alpha_d$  itself must be a rational multiple of  $\pi$ , and by Niven's theorem, since  $\cos \alpha_d$  is rational, this happens only for  $d \in \{1, 2, 4\}$ . So a pair whose  $d$  is alone in its class, among the values of  $d$  that occur, can never be a rim neighbour pair.

*Proof of Theorem 1 for  $n = 5$ .* The values of  $d$  for  $2 \leq j \leq k \leq 5$  are 1, 2, 3, 4, 6, 8, 9, 12, 16, and 6, 8, 12, 16 are alone in their classes 5, 7, 11, 15. So none of the pairs  $(\frac{1}{3}, \frac{1}{4})$ ,  $(\frac{1}{3}, \frac{1}{5})$ ,  $(\frac{1}{4}, \frac{1}{5})$ ,  $(\frac{1}{5}, \frac{1}{5})$  can be neighbours on the rim, and a  $\frac{1}{5}$  coin must sit between two half coins, at  $\alpha_4 = 60^\circ$  from each. Those two half coins are then  $120^\circ$  apart, with centres  $2 \cdot \frac{1}{2} \sin 60^\circ = \frac{\sqrt{3}}{2} < 1$  apart, so they overlap (Figure 1(c)).  $\square$

#### 4. ALL RIM ARRANGEMENTS UP TO $n = 1000$

The same reasoning, done by computer, decides every  $n$ . Since every multiplicity in (2) is nonnegative, there is a cheap first step.

*Pruning.* In a class, if at some split prime every remaining  $d$  with a nonzero valuation difference has the same sign, then (4) forces all of them to have multiplicity 0; remove them and repeat. If some size  $\frac{1}{j}$  is left with no possible neighbour, there is no rim arrangement.

For  $7 \leq n \leq 10$  and  $20 \leq n \leq 1000$  the pruning alone strands some size, for example  $\frac{1}{7}$  for  $7 \leq n \leq 10$  and  $\frac{1}{20}$  for  $20 \leq n \leq 27$ ; the script prints the stranded sizes for each  $n$ . For the remaining values  $n = 5, 6$  and  $11 \leq n \leq 19$  an exhaustive search finishes the job:

- (1) in each class, list the multiplicity vectors with angle sum at most  $2\pi$  that satisfy (4), and their exact values;
- (2) combine the classes in all ways that give exactly  $2\pi$ ;
- (3) assign each  $d$  to pairs  $(j, k)$  with  $(j - 1)(k - 1) = d$ , keep the multigraphs on the sizes that use every size and have even degrees, and search their Euler circuits for one in which no two coins overlap.

The search reproduces the known small cases (three cycles for  $n = 3$ , among them the asker's, and exactly the asker's ring for  $n = 4$ ) and finds nothing for  $5 \leq n \leq 22$ ; at  $n = 22$  there are 1264 multisets of angles summing to exactly  $2\pi$ , and none survives step 3. Every overlap that rejects a candidate does so by a margin of at least 0.05 in the radius-1 scale, far above rounding error, and no search was truncated. This proves Theorem 1.  $\square$

#### 5. CORONAS AND TRIANGULATED ARRANGEMENTS

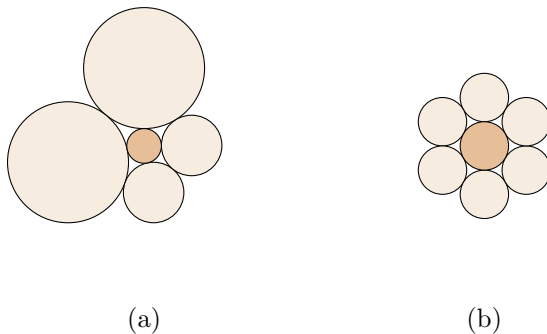
The bookkeeping works around any coin, not only around the centre of the tray. Let a coin of curvature  $m$  touch two bodies  $a$  and  $b$  that touch each other, where the tray counts as curvature  $-1$ . The triangle of centres has sides  $|\frac{1}{m} + \frac{1}{a}|$ ,  $|\frac{1}{m} + \frac{1}{b}|$ ,  $|\frac{1}{a} + \frac{1}{b}|$ , so the angle between the two contact directions has rational cosine  $c = p/q$  (with the sign of  $c$  reversed when one of the bodies is the tray, whose contact direction points away from its centre). With  $q^2 - p^2 = s^2 a$  this angle is  $\arg(p + s\sqrt{-a})$ , and the splitting and valuation criteria apply unchanged.

A *closed corona* of a coin is a cyclic list of neighbours, each touching the next, with no two overlapping, the tray used at most once, and contact directions turning by exactly  $2\pi$ . Coronas of this kind, with exact angle relations, are the main tool in the classification of compact packings of the plane by discs of two and three sizes [7, 5]. If the contact graph of an arrangement, tray included, is a triangulation, then every coin has a closed corona, namely its actual neighbours in cyclic order, and the contact graph is connected.

*Proof of Theorem 2.* Let  $R$  be the smallest set of sizes containing every size with a closed corona that uses the tray, and every size with a closed corona that uses a size in  $R$ . In a triangulated arrangement, follow a path of contacts from any coin to the tray; walking it backwards shows that every size present lies in  $R$ . Computing all closed coronas exactly, with the criteria above, gives the sizes outside  $R$ :

| $n$        | 5–7           | 8                          | 9                                       | 10, 11                     | 12            | 13–16                       |
|------------|---------------|----------------------------|-----------------------------------------|----------------------------|---------------|-----------------------------|
| not in $R$ | $\frac{1}{5}$ | $\frac{1}{5}, \frac{1}{8}$ | $\frac{1}{5}, \frac{1}{8}, \frac{1}{9}$ | $\frac{1}{8}, \frac{1}{9}$ | $\frac{1}{9}$ | $\frac{1}{9}, \frac{1}{13}$ |

So for every  $5 \leq n \leq 16$  some size cannot occur. The angle relations in this computation are decided exactly; the overlap tests are numerical, but a candidate is rejected only when two coins overlap by more than  $10^{-6}$ , and every rejection in fact has a margin of at least 0.004.



**Figure 2.** Two closed coronas. (a) A  $\frac{1}{7}$  coin ringed by  $\frac{1}{2}, \frac{1}{2}, \frac{1}{4}, \frac{1}{4}$ : the four angles between consecutive contacts add up to exactly  $2\pi$ , an identity the valuation test certifies. (b) Six  $\frac{1}{5}$  coins around a  $\frac{1}{5}$  coin, the only closed corona of a  $\frac{1}{5}$  coin when  $5 \leq n \leq 7$ .

Near-tangencies are accepted, which can only add coronas and enlarge  $R$ , and no search was truncated.  $\square$

For  $5 \leq n \leq 7$ , for example, the only closed corona of a  $\frac{1}{5}$  coin is six  $\frac{1}{5}$  coins (Figure 2(b)), so in a triangulated arrangement the  $\frac{1}{5}$  coins would form a region that never reaches the tray. The computation does find other exact coronas, such as a  $\frac{1}{7}$  coin ringed by  $\frac{1}{2}, \frac{1}{2}, \frac{1}{4}, \frac{1}{4}$  (Figure 2(a)), which is the identity behind the asker's near-fit with  $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{6}, \frac{1}{7}$ .

## 6. COINS IN A CORNER: A CONGRUENCE

*Proof of Proposition 3.* If the tray (curvature  $-1$ ) and coins of curvatures  $b, c, k$  are mutually tangent, Descartes' circle theorem holds for the quadruple  $(-1, b, c, k)$  of integers, and the Apollonian packing it generates is integral and contains the tray. The theory of root quadruples [6] reduces every integral Apollonian packing to a unique root quadruple  $(a, b, c, d)$ , one with  $a < 0 \leq b \leq c \leq d$  and  $a + b + c \geq d$ , where  $a$  is the curvature of the bounding circle. Here  $a = -1$ , and  $b \geq 2$  because every other circle of the packing is smaller than the tray. Put  $x = b + c - 1$ . Descartes' equation  $(x + d)^2 = 2(1 + b^2 + c^2 + d^2)$  rearranges to  $(x - d)^2 = 4(bc - b - c)$ , while  $0 \leq x - d \leq b - 1$  because  $c \leq d \leq x$ . Since  $c \geq b$ ,  $bc - b - c \geq b^2 - 2b$ , so  $4(b^2 - 2b) \leq (b - 1)^2$ , which forces  $b = 2$ ; then  $4(c - 2) \leq 1$  gives  $c = 2$ , and  $(x - d)^2 = 0$  gives  $d = 3$ . So the root quadruple is  $(-1, 2, 2, 3)$ . By the congruence theorem of Graham, Lagarias, Mallows, Wilks and Yan [6], the curvatures in that packing lie in the classes  $2, 3, 6, 11, 14, 15, 18, 23 \pmod{24}$ ; a direct computation of all its curvatures up to 5000 agrees.  $\square$

So the familiar exact constructions (filling the gap between the tray and two touching coins, Pappus chains, Steiner chains around a central coin, which inversion turns into Descartes quadruples) can never produce a  $\frac{1}{4}$ ,  $\frac{1}{5}$  or  $\frac{1}{7}$  coin. Every arrangement with  $n \geq 4$  must lock its  $\frac{1}{4}$  coin some other way; the asker's ring does it with the identity  $2\alpha_3 + \alpha_9 = \pi$ .

*Remark 4* (Why searches find near-misses; heuristic). With  $N$  coins there are  $2N - 1$  degrees of freedom, and a coin placed against two existing bodies uses up exactly its two. Rigidity needs a positive self-stress, which generically needs at least one contact more than the degrees of freedom; a configuration already determined meets an extra tangency only through an algebraic coincidence. So numerical searches produce endless near-packings, and an exact arrangement, if one exists, rests on an identity such as  $2\alpha_3 + \alpha_9 = \pi$ . The rim case is exactly where such identities can be classified completely.

## 7. VERIFICATION

The law-of-cosines formula (1), the fact that  $\alpha_d$  is a rational multiple of  $\pi$  only for  $d \in \{1, 2, 4\}$  (from Niven's theorem in Mathlib), the identity  $2\alpha_3 + \alpha_9 = \pi$ , the overlap in the  $n = 5$  argument and the uniqueness of the root quadruple  $(-1, 2, 2, 3)$  are proved in Lean 4 [8]. The splitting theorem [4], the valuation criterion (standard algebraic number theory, as explained in

Section 3) and the congruence theorem [6] are quoted, and the searches are computations. The valuation criterion is cross-checked against 60-digit numerics on 122,400 random combinations (944 relations, no disagreement), and deleting one prime from the valuation vectors makes that check fail. The rim and corona searches report truncations, the smallest overlap margins and near-tangencies, and planted errors (a numerical relation test with tolerance  $10^{-3}$  for the rim,  $2 \cdot 10^{-2}$  for coronas) change their output. Code and Lean files are at [9].

#### ACKNOWLEDGMENTS

The question and the arrangements for  $n \leq 4$  are Dan's. The reduction of the rim case to the angles  $\alpha_d$  is from the answer [2], and the near-miss searches are Matthew House's [3]. The analysis, the computations and the Lean formalisation were carried out with Claude Opus 5.5 (Anthropic). The author is responsible for the content.

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